Software Engineering

SW Development Process & models

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Agenda

- ✓ **Motivation**
  - Risks
  - Famous SW bugs
- ✓ **Process definition**
  - How is a model used?
  - Why is a model important?
- ✓ **Development models**
  - Spiral model
  - Waterfall model
  - V model (V-Cycle)
- ✓ **Summary**



Motivation



Risks:

- Software Bugs may have dramatic consequences
- Software Bugs are systematic i.e. in every product!
- Supplier pays for recall costs
- Product Liability
- SW complexity often underestimated





Motivation



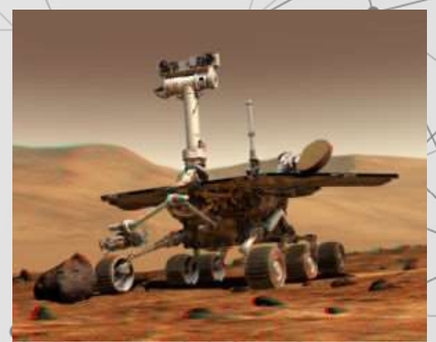


Famous SW bugs:

- Software Bugs may have dramatic consequences
  NASA Apollo 11 landing problem (July 20, 1969).
- ESA Ariane 5 Flight 501 self-destruction 40 seconds after takeoff (June 4, 1996).



- NASA Mars Climate Orbiter destroyed due to entry of momentum data in imperial units instead of the metric system (September 23, 1999).
- NASA Mars Rover freezes due to too many open files in flash memory (January 21, 2004).



- NASA Mars Global Surveyor battery failure was the result of a series of events linked to a computer error made five months before (November 2, 2006).

Motivation

Famous SW bugs:



- The Therac-25 accidents (1985-1987), which caused at least five deaths.
- A misuse of medical diagnosis software created by Multidata Systems International, at the National Cancer Society in Panama City, caused, by different estimates, between five and eight cancer patients to die of over-radiation. (2000)
- The year 2000 problem, popularly known as the "Y2K bug", spawned fears of worldwide economic collapse and an industry of consultants providing last-minute fixes.



General Definitions

How do you define process?

- A process is a **set of practices** performed **to achieve a given purpose**; it may include tools, methods, materials, and/or people.

- While process is often described as a leg of the process-people-technology triad, it may also be considered the **“glue” that unifies** the other aspects.



What is a process model?

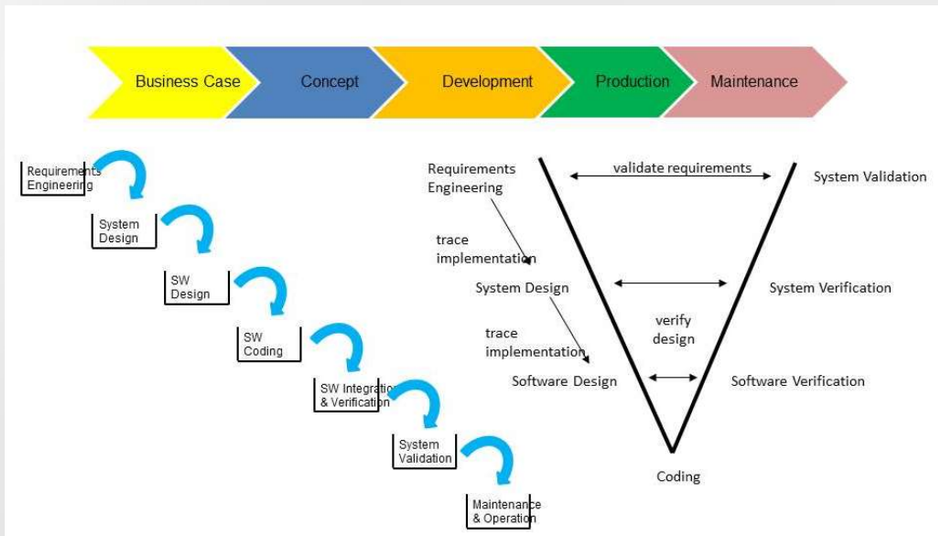
- A model is a **structured collection** of elements that **describe characteristics** of effective processes.
  - Processes included are those proven by experience to be effective.



How is a model used?

A model is used…

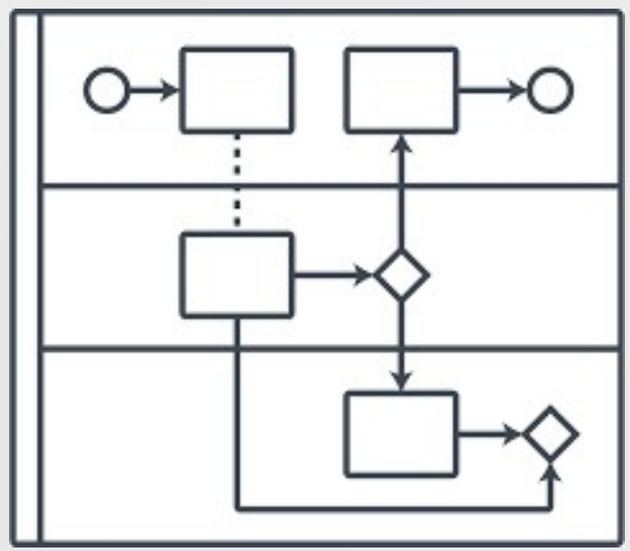
- to help set process improvement objectives and priorities, improve processes, and provide guidance for **ensuring stable**, capable, and mature processes
  - as a **guide for improvement of organizational** processes



Why is a model important?

A model provides…

- a place to start
- the benefit of a community's prior experiences
- a common language and a shared vision
- a framework for prioritizing actions

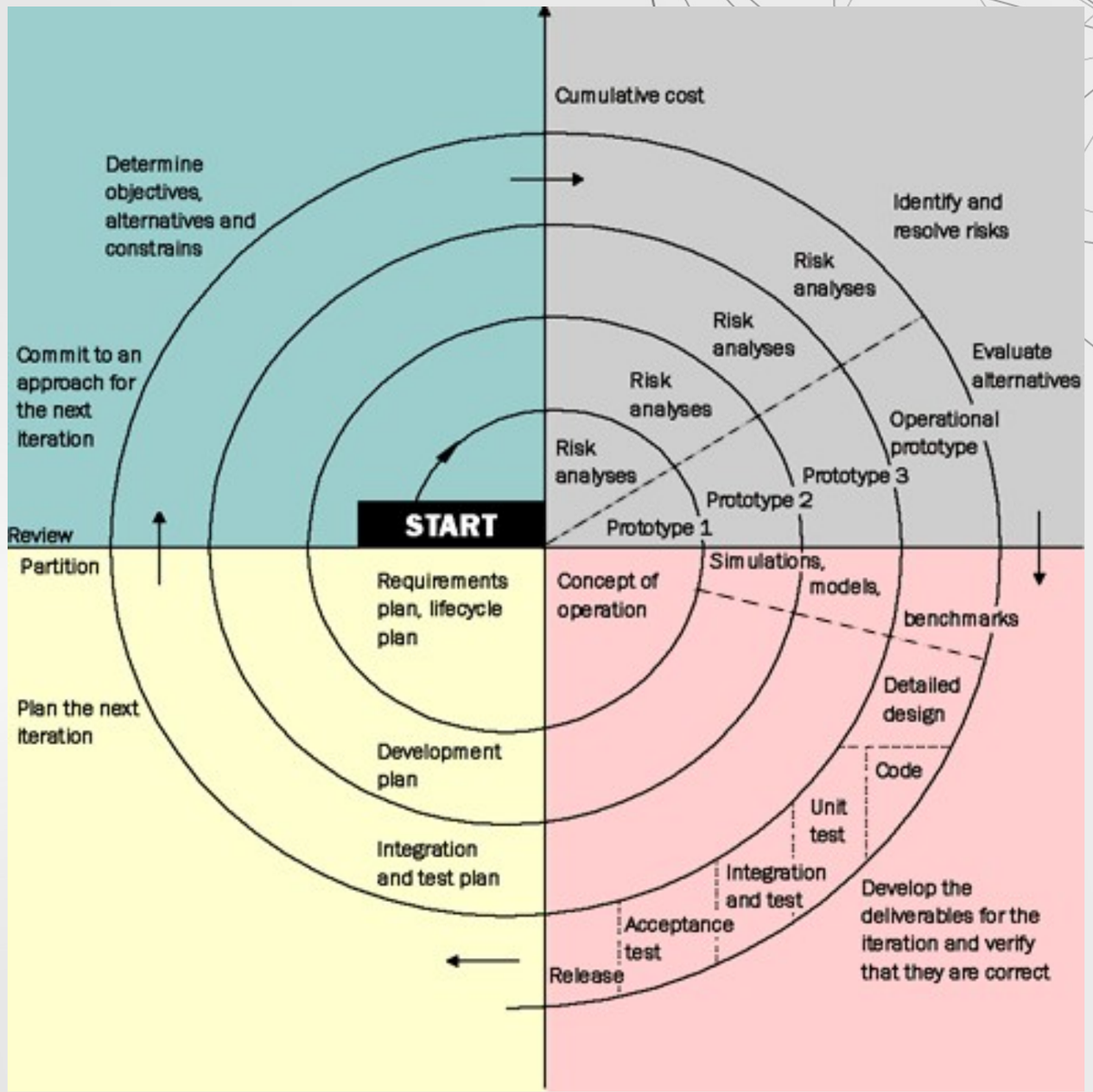


Development models



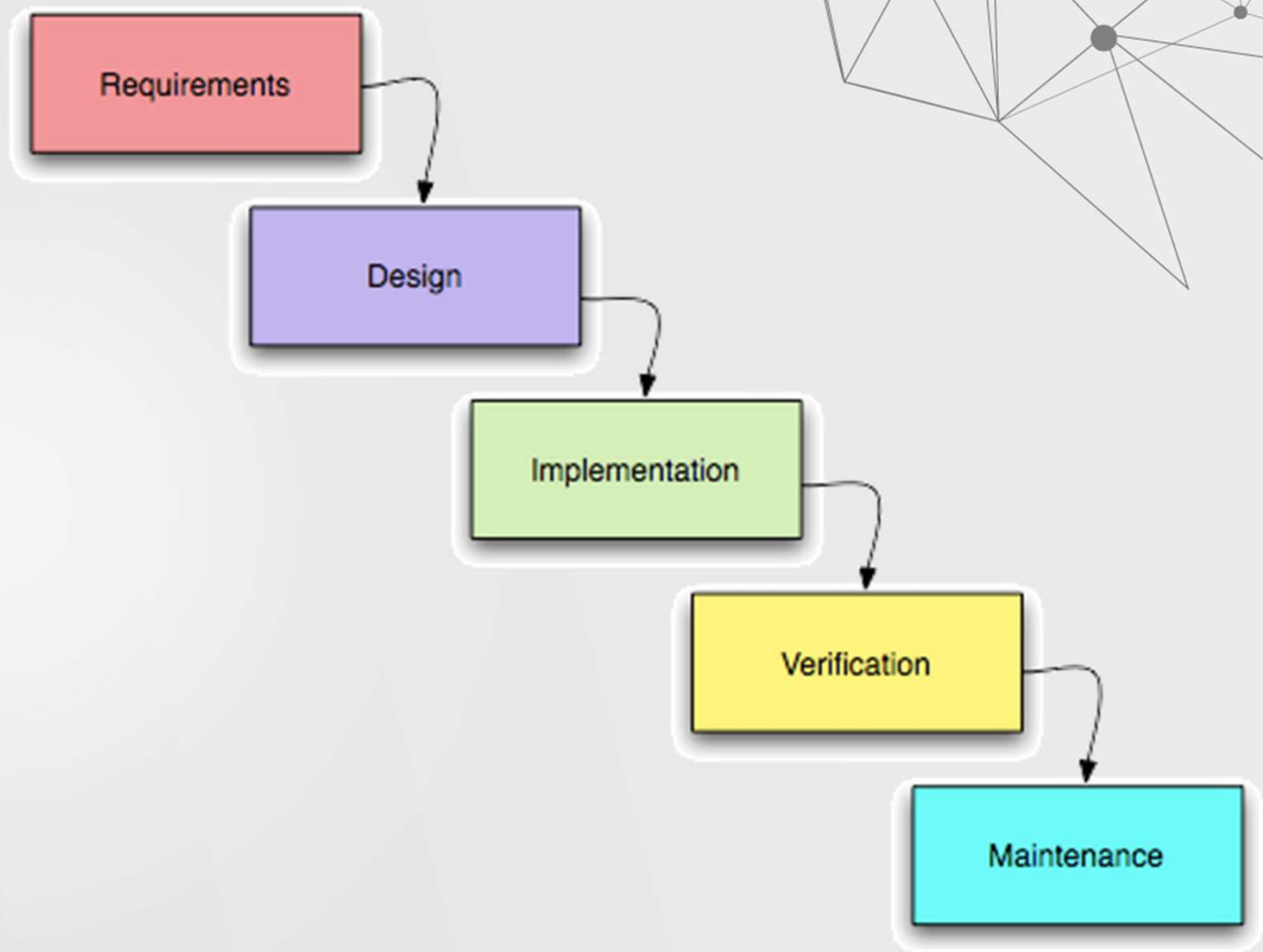
Spiral Model:

- Prototypes get more details as project advance
- Radial dimensions suggests project cost
- Angular dimension represents progress within each iteration
- Allows changes to specifications
- Planning and estimations can get realistic due to experience gain in previous iterations
- People workload is more homogenous.



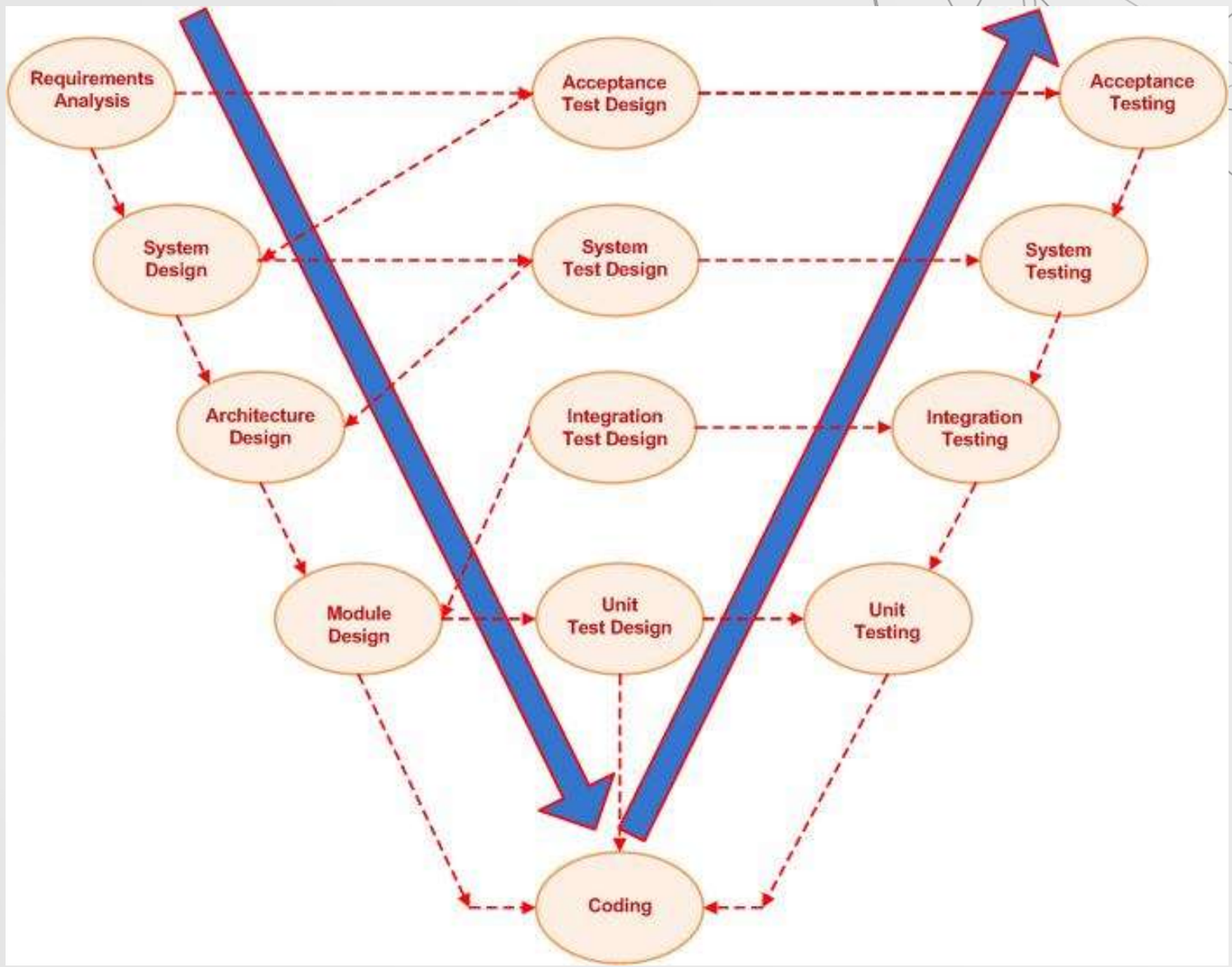
Waterfall Model:

- Development phases start after the previous phase ended
- Documentation is very important
- Requirements have to be frozen before design is started
- Verifications are embedded in each phase
- **No changes possible**
- Management control is difficult
- After one phase is over, the expert have not work and have to be assigned for other projects

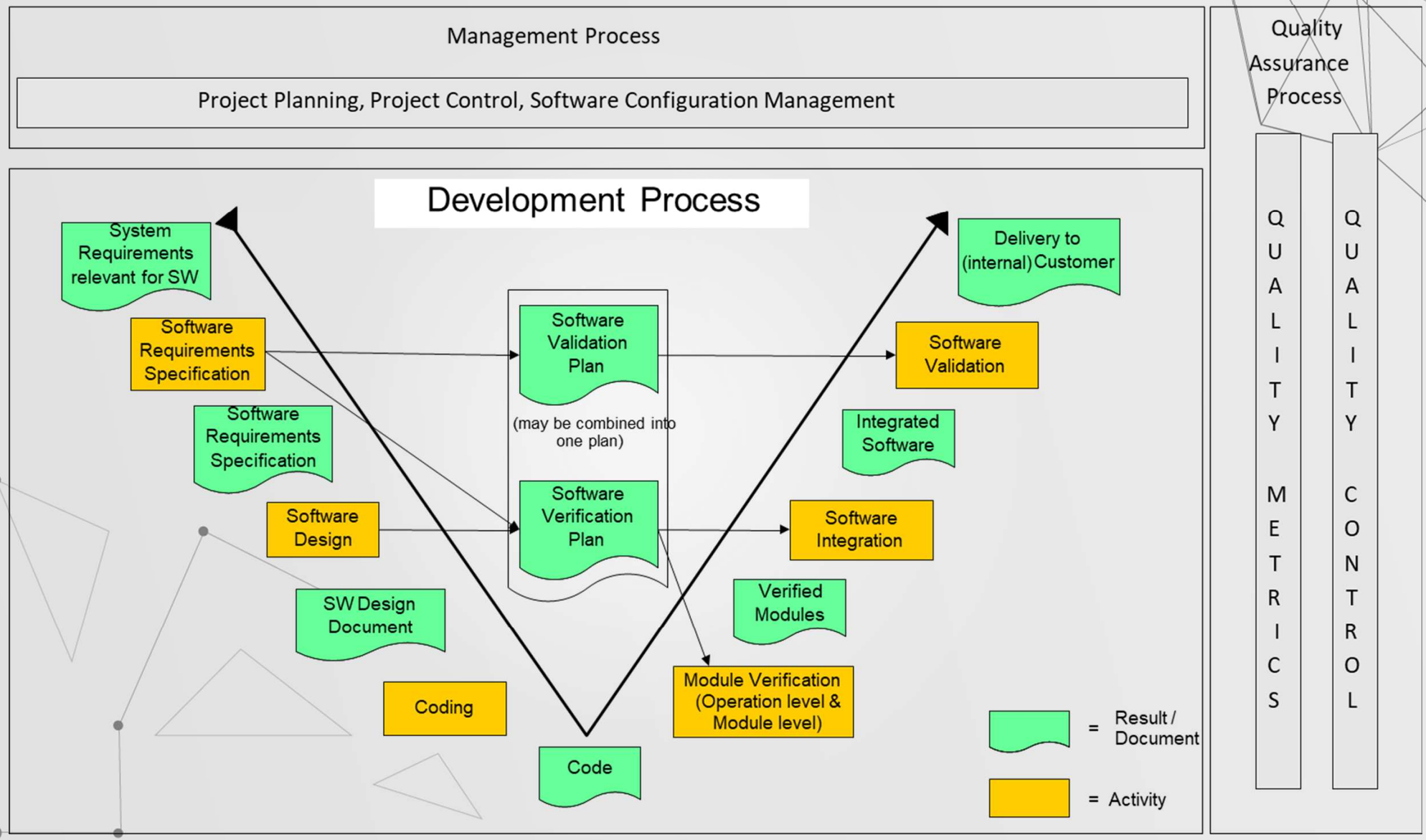


V Model or “V Cycle”:

- V model is and extension of the Waterfall model
- V model emphasize the relations between the development phases and associated verification phases

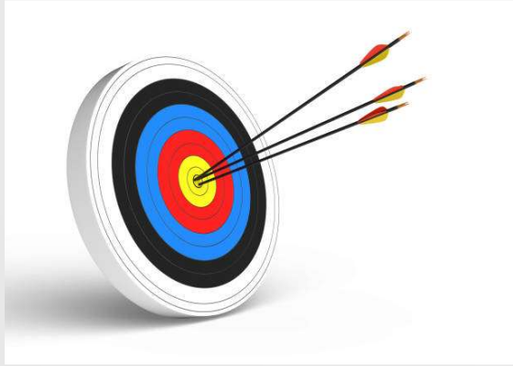


SW Development V-Cycle



Summary

- Aims of the procedure:
  - Concentration on what has to be done to develop software of high quality in time, not on how it should be done.
  - **Satisfy the needs** and requirements expressed by the customer
  - Create **software products in a repeatable manner** based on a defined process
  - **Reduce the number of defects** and the costs to correct them
  - **Guarantee the agreed time frame**, cost framework and results via a controlled project progress



Questions?

